

GuideVoice

A conversational walking guide for blind and low-vision users.

The Idea

GPS apps tell you “*turn left in 200 feet*” but not that there’s a construction barrier, which door is the entrance, or where the crosswalk button is. GuideVoice combines turn-by-turn directions with the phone camera: point it ahead and ask “*What’s in front of me?*” or “*Where’s the door?*”, and Grok Voice answers in a calm, conversational way. This directly answers an opening ceremony prompt: “*What types of audio or haptic navigation would blind or low-vision users benefit from?*”

Track Fit

Built with Cursor	Whole app built in Cursor.
Grok Voice / Imagine	Voice is the entire interface.
Space data	Not used. Allowed by the rules, but the track is pitched around space, so judges may weigh this.
Navigation theme	Direct fit, and an accessibility prompt from the slides.
Also eligible	Big Red, Software, Design, People’s Choice.

Features

MVP

- Voice-only start: “*Take me to Klarman Hall.*”
- Walking directions spoken at each turn.
- “What’s ahead?” button: snap a photo, get a short spoken description focused on obstacles, doors, and signs.
- Large-text, high-contrast, screen-reader-friendly UI.

Stretch

- Haptic patterns for left/right/arrived (phone vibration).
- Continuous mode that describes changes every few seconds.
- Read signs and room numbers aloud.

Tech Stack and Data

- Frontend: mobile web app (PWA) with camera and geolocation.
- Routing: Google Maps Directions API (workshop Saturday 1:00 PM) or OpenStreetMap routing.
- Vision: an image-understanding model (Grok’s vision-capable chat model, to verify).
- Voice: Grok Voice API.

Limitations and Risks

- **Safety.** Wrong or late descriptions could lead a blind user into danger. It must never replace a cane, guide dog, or orientation training.
Mitigation: Position it as an assistant, include a clear disclaimer, and never say “it’s safe to cross.”
- **Latency.** Photo upload, AI description, and speech take seconds; a walking person moves several meters meanwhile.
Mitigation: Keep descriptions short; use the on-demand button instead of continuous mode for the MVP.
- **GPS accuracy.** Phone GPS is off by 5–20 meters near buildings, which matters a lot to someone who cannot see.
Mitigation: Use the camera to confirm landmarks (“the entrance is on your right”).
- **No real users to test with.** Without blind or low-vision testers, design choices are guesses.
Mitigation: Follow established accessibility guidelines; the Saturday 3:00 PM Assistive Tech workshop may help.
- **Existing products.** Be My AI, Seeing AI, and Google Lookout already describe scenes.
Mitigation: Differentiate by combining scene description *with* route guidance in one conversation.
- **Grok API unknowns.** Image-input support and Voice API details are not verified.
Mitigation: Confirm at the Friday Grok/Cursor workshop.

Scorecard

Demo-able	5/5	Works live, indoors, at the judging table.
Buildable in 24h	4/5	Mostly API integration.
Navigation theme	5/5	Direct accessibility prompt.
Wow factor	4/5	Judges can close their eyes and try it.
Technical risk (5 = riskiest)	2/5	Low to moderate.

Verdict

Top non-space pick. Meaningful, demos live indoors, and matches a prompt from the opening slides. Be careful with safety claims.